Alejandro Ciuba

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Education

University of Pittsburgh August 2023 – Present
Ph.D. in Computer Science Pittsburgh, PA
- **Research:** Community-focused natural-language processing GPA: 3.91
- **Advisors:** Aakash Gautam and Lorraine (Xiang) Li

University of Pittsburgh August 2019 – May 2023
B.S. in Computer Science Pittsburgh, PA
Minors in Linguistics and Spanish GPA: 3.97

Research Experience

An Exploration of Graduate Student LLM Usage and Attitudes January 2026 – Present
Seminar Project under Dr. Lorraine (Xiang) Li Pittsburgh, PA
- Designing large-scale survey targeting graduate students on their LLM usage and attitudes on LLM ethics
- Utilizing interviews and contextual inquiry to document graduate student workflows with and without LLMs

Game Design for Family-Focused Language Promotion August 2025 – Present
Project Leader Pittsburgh, PA
- Creating a family-focused language learning game aimed at heritage speakers and their parents
- Supervising two undergraduate students on analysis, design, at-home user studies and writing phases

Low-Resource Language Preservation and Community Engagement January 2024 – Present
Graduate Student Researcher under Dr. Aakash Gautam Pittsburgh, PA
- Developing a literature analysis of NLP community engagement practices using the PRISMA framework
- Conducting interviews and thematic analysis to better understand how technology can benefit communities

Thematic Analysis of User Survey Responses on Social Media Bans October 2024 – June 2025
Graduate Student Researcher under Dr. Aakash Gautam Pittsburgh, PA
- Conducted a thematic analysis of survey responses regarding social media in Nepal
- Synthesized response views into themes to discuss social media policies

Automatic Essay Evaluation Using Linguistic Features May 2024 – November 2024
Graduate Student Researcher under Dr. Lorraine (Xiang) Li Pittsburgh, PA
- Performing baseline training/fine-tuning on a custom BERT-based model via PyTorch and HuggingFace
- Conducting exploratory data analysis on essay corpora via feature correlation and hypothesis testing

ASR Transcription Research & Technical Support August 2022 – August 2024
Research Assistant under Dr. Dan Villarreal Pittsburgh, PA
- Developed custom ASR pipeline with PyTorch Lightning and `pyannote` to automatically segment audio
- Developed a Python API to work with ELAN files with `pytest` integration

Classroom Discussion Analysis Research January 2023 – April 2023
Research Assistant under Dr. Diane Litman Pittsburgh, PA
- Managed dataset cleaning pipeline for Word Documents, focusing on anonymization and XML tag extraction

- Examined teacher-student classroom discussion differences through data analysis and hypothesis testing

Frederick Jelinek Memorial Summer Workshop June 2022 – August 2022
Undergraduate Researcher under Drs. Anthony Larcher and Santosh Kesiraju Baltimore, MD
- Created a trilingual corpus (English, French and Tamasheq/Tamahaq) for low-resource machine translation
- Ran low-resource model training simulations (e.g. for T5-Small and mBART)

Linguistics Research January 2022 – April 2022
Undergraduate Research Assistant under Drs. Alan Juffs and Na-Rae Han Pittsburgh, PA
- Parsed video game scripts for pragmatics analysis using `spaCy`, `nltk` and `beautifulsoup`
- Aided in the cleaning of the unpublished PELIC Speech Dataset

Work Experience

UR2PhD Mentorship Training September 2025 – December 2025
Undergraduate researcher mentorship course through the Computing Research Association Pittsburgh, PA
- Mentors an undergraduate student one-on-one research with the goal of publishing
- Guides the student through the research process, including data analysis, literature review, and paper writing
- Scaffolds tasks for the student to facilitate their learning experience
- Engages with the student via weekly meetings to discuss progress and develop tasks to complete together

NOUR Mentor May 2025 – August 2025
Undergraduate researcher mentor Pittsburgh, PA
- Guided undergraduate students in their first research experience
- Conducted lessons on research know-how (e.g., citation management, data analysis, problem discovery).
- Helped students present their findings in a final poster presentation at the end of the course.

Teaching Assistant Positions August 2021 – Present
Computational Linguistics and Data Structures & Algorithms Pittsburgh, PA
- Ran lab sessions for students, covering fundamental algorithm concepts
- Hosted office hours for students learning computational linguistic and natural-language processing concepts
- Conducts lab session for students, covering web design fundamentals
- Hosts office hours sessions to work one-on-one with students
- Scaffolds problems for students to develop their problem-solving skills

Carnegie Library ESL Class August 2019 – March 2020
Volunteer ESL Instructor for the Carnegie Public Library Pittsburgh, PA
- Guided adult learners through vocabulary and grammar exercises
- Maintained a safe environment for English learners

Publications

CHI 2026

1. **Alejandro Ciuba**, Zheng YY Li, and Aakash Gautam. "Not Just Duolingo: Supporting Immigrant Language Preservation Through Family-Based Play". en. In: (Mar. 2026). arXiv:2604.00282 [cs]. DOI: `10.48550/arXiv.2604.00282`. URL: `http://arxiv.org/abs/2604.00282`

CSCW 2025

2. Prerana Khatiwada, **Alejandro Ciuba**, Aditya Nayak, Aakash Gautam, and Matthew Louis Mauriello. "Regulating Social Media: Surveying the Impact of Nepali Government's TikTok Ban". In: *Proc. ACM Hum.-Comput. Interact.* 9.7 (Oct. 2025), CSCW467:1–CSCW467:33. DOI: `10.1145/3757648`. URL: `https://dl.acm.org/doi/10.1145/3757648`

AAAI iRaise Workshop

3. Zhaoyi Hou, **Alejandro Ciuba**, and Xiang Li. "Improving LLM-based Automatic Essay Scoring with Linguistic Features". In: *Proceedings of the Innovation and Responsibility in AI-Supported Education Workshop*. Ed. by Zichao Wang, Simon Woodhead, Muktha Ananda, Debshila Basu Mallick, James Sharpnack, and Jill Burstein. Vol. 273. Proceedings of Machine Learning Research. PMLR, Mar. 2025, pp. 41–65. URL: `https://proceedings.mlr.press/v273/hou25a.html`

INTERSPEECH 2023

4. Santosh Kesiraju, Marek Sarvaš, Tomáš Pavlíček, Cécile Macaire, and **Alejandro Ciuba**. "Strategies for Improving Low Resource Speech to Text Translation Relying on Pre-trained ASR Models". In: *INTERSPEECH 2023*. ISCA, Aug. 2023. DOI: `10.21437/interspeech.2023-2506`

Horizons 2020 Project

5. Anthony Larcher, Yannick Estève, Mickael Rouvier, … Cecile Macaire, **Alejandro Ciuba**, Peter Vickers, Luis Vicente, Victoria Mingote, and Pablo Gimeno. *Multi-Lingual Speech to Speech Translation for Under-Resourced Languages*. Tech. rep. Le Mans Université, Aug. 2022

Invited Presentations

Human-Computer Interaction + Natural-Language Processing March 2025
Python Linguistics Club

Guest Lecture on Machine Translation December 2023
Introduction to Computational Linguistics

Grants & Fellowships

Computing Research Association UR2PhD September 2025 – Present
Amount: *$1,000 USD*

Southern Regional Educational Board Fellowship August 2024 – Present
Amount: *$750 USD in travel funds for three years*

K. Leroy Irvis Fellowship August 2023 – Present
Amount: *$30,000 USD*

Awards

CSC Hacks Winner + Best Beginner Project November 2020
Submission: *Flatland: The Game*

Provost Academy Member August 2019 – May 2023

Skills

Programming & Computation

- C, Python, R, Go, Java
- Git, GitHub and GitLab
- Compute cluster usage via Slurm
- Linux, Docker and `bash` script creation

User Research Skills

- Quantitative analysis via inferential statistics
- Qualitative analysis via coding and thematic analysis
- User interview and survey creation

Foreign Languages & Linguistics

- Conversational Spanish
- Basic Portuguese Comprehension
- Manual document translation experience